SOFTWARE REQUIREMENTS SPECIFICATION

SMARTMED

Medication Management & Adherence Monitoring System

IEEE-style SRS | Version 1.0 | Academic Session 2026–27

SUBMITTED BY	INSTITUTION
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Document Control

Field	Details
Document title	Software Requirements Specification (SRS) — SmartMed
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System type	Web-based medication management and adherence monitoring platform
Source basis	SmartMed_IEEE_Research_Paper.pdf and NEW SMARTMED project report.pdf supplied for this task
Intended audience	Project evaluator, mentor, developers, testers and academic reviewers
Conformance note	Structured in an IEEE-style SRS format; requirements are based on the supplied documents. Unspecified implementation details are explicitly identified rather than assumed.

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1. Introduction

1.1 Purpose

This Software Requirements Specification defines the intended behaviour, interfaces, data rules, security boundaries and verification criteria for SmartMed. It provides a common, testable reference for implementation, demonstration and evaluation. A requirement is considered satisfied only when its stated behaviour can be demonstrated or verified.

1.2 Scope

SmartMed is a browser-based application for organizing patient medication records, schedules, scheduled-dose events, adherence history and summary analytics. It supports patient-controlled invitations to caregivers and doctors for read-only monitoring, recipient-scoped in-app notifications, and an informational medication-interaction workflow.

The system is an educational software project for personal organization and demonstration. It is not a medical device; it does not diagnose a condition, prescribe treatment, issue emergency alerts, or guarantee that a medicine combination is safe.

1.3 Intended Audience

- Project evaluators and academic reviewers assessing requirement completeness and testability.
- Developers implementing the React/TypeScript frontend and Java/Spring Boot backend.
- Testers verifying functional behaviour, access control, data integrity and error handling.
- Project members maintaining documentation and planning future improvements.

1.4 Definitions and Abbreviations

Term	Definition
SRS	Software Requirements Specification.
JWT	JSON Web Token used as a bearer access token.
RBAC	Role-Based Access Control.
Patient	Account that owns medication records and controls care-team consent.
Caregiver / Doctor	Invited account allowed to view permitted patient monitoring data only while a matching relationship is ACTIVE.
Dose record	A saved occurrence of a scheduled medication dose, including its date, time and application status.
Soft deactivation	Marking a medication/schedule inactive while preserving historical dose records.
NO_DATA	Interaction-check response indicating that no interaction information was returned; it does not mean a combination is safe.
Flyway	Database migration tool used to version schema changes.
API	Application Programming Interface.
IANA timezone	Named timezone identifier; SmartMed's documented default is Asia/Kolkata.

1.5 References and Source Basis

- SmartMed: A Consent-Driven, Layered Web Platform for Medication Management and Adherence Monitoring Using React, Spring Boot and MySQL (supplied IEEE-style research paper).
- NEW SMARTMED project report (Foundation Course, academic session 2026–27).

Where the two source documents differ in wording or level of detail, this SRS consolidates the shared system behaviour and preserves explicit limitations. It does not claim new implementation evidence, performance measurements or clinical validation.

2. Overall Description

2.1 Product Perspective

SmartMed uses a three-tier architecture. The browser-based presentation layer communicates with a versioned REST API; the backend validates requests and enforces authentication, authorization and business rules; repositories persist data in MySQL. Flyway migrations are the authoritative mechanism for schema evolution.

Layer	Technology / component	Primary responsibility
Presentation	React 18, TypeScript, Vite 5, Chart.js	Forms, navigation, dashboards, date-based views and charts.
API client	Typed frontend API modules	Serialize requests, attach bearer tokens, process success/error responses.
Backend	Java 17, Spring Boot 3, Spring Web, Spring Data JPA	REST endpoints, domain logic, transactions and persistence.
Security	Spring Security, JWT, BCrypt, validation and rate limiting	Authentication, role/ownership checks, credential protection and request validation.
Database	MySQL 8, Flyway	Relational records, foreign keys, unique constraints and versioned schema.
Testing/build	Maven wrapper, npm, JUnit, Spring Boot Test, H2	Build and automated test workflows.

2.2 Product Functions

- Register a patient account and authenticate users.
- Display the authenticated user's profile without credential secrets.
- Create, view, update and deactivate patient-owned medication records.
- Maintain schedules and create dose records without duplicates.
- Record dose outcomes and retain a queryable history.
- Calculate date-filtered adherence summaries from saved dose states.
- Invite, accept, decline and revoke caregiver/doctor relationships.
- Provide read-only dashboards to connected care-team users with active matching consent.
- Request an informational interaction check through a pluggable provider interface.
- Create and manage recipient-owned in-app notifications.

2.3 User Classes and Characteristics

Actor	Responsibilities and permitted use
Patient	Owns medication data; manages schedules and dose outcomes; views own analytics and notifications; invites and revokes care-team access.
Caregiver	Accepts or declines a patient invitation; views only permitted monitoring information for patients with an ACTIVE CAREGIVER relationship; cannot change patient dose status.
Doctor	Accepts or declines a patient invitation; views permitted monitoring information for patients with an ACTIVE DOCTOR relationship; cannot change patient dose status through monitoring routes.
System services	Validate tokens and requests, enforce current database roles and relationships, generate dose events, maintain notifications and apply migrations.

2.4 Operating Environment

Category	Requirement
Client OS	Windows, Linux or macOS.
Browser	Modern browser with JavaScript enabled.
Frontend	Node.js 18+ and npm for development/build.
Backend	JDK 17+ and Maven wrapper.
Database	MySQL 8 or a compatible configured instance.
Development hardware	Modern dual-core processor or better; 8 GB RAM recommended.
Network/API	REST over JSON; HTTPS for deployed environments; authenticated endpoints use Bearer access tokens.

2.5 Design and Operating Constraints

- Public registration creates a PATIENT account only; users cannot assign themselves caregiver or doctor roles through a registration field.
- All protected authorization decisions are server-side and based on the authenticated account currently loaded from the database.
- Care-team access is read-only and requires an ACTIVE relationship of the matching type.
- Medication deactivation preserves dose history; it is not a physical deletion of historical data.
- The default interaction provider is a mock provider with no real clinical interaction dataset.
- Database schema changes are managed by Flyway; Hibernate validates rather than automatically creating/updating the schema.
- Secrets, database settings, CORS origins, TLS configuration and timezone are environment/configuration dependent.

2.6 User Documentation

The project report describes patient, caregiver and doctor workflows. For a complete submission, the final project package should include setup instructions, environment configuration guidance, role-based workflow guidance and the safety note that interaction results and adherence percentages are informational only.

3. System Features and Functional Requirements

The identifiers below are stable references for implementation and testing. Priority is expressed as Core or Supporting; it does not imply that unimplemented features exist.

ID	Feature	Priority	Requirement statement	Acceptance condition
FR-01	Patient registration	Core	The system shall accept a full name, email address and password for public registration. It shall normalize the email, enforce uniqueness and create a PATIENT account regardless of any client-supplied role field.	A valid unique registration creates one patient account; duplicate/invalid data is rejected with a structured error; no password or hash is returned.
FR-02	User authentication	Core	The system shall verify login credentials and issue an access token and user summary only when credentials are valid. Invalid login responses shall use a generic invalid-credentials message.	Valid credentials authenticate; invalid credentials do not issue a token or reveal whether an email is registered.
FR-03	Current profile	Core	The system shall return the authenticated user's own profile through a protected endpoint and shall exclude password fields, password hashes and token secrets.	Unauthenticated requests are denied; response contains the current account profile without credential secrets.
FR-04	Medication management	Core	The system shall allow an authenticated patient to create, view, update and deactivate medication records owned by that patient. The server shall derive ownership from the authenticated principal.	Owner operations succeed when valid; attempts to access another patient's medication are rejected; deactivation retains history.
FR-05	Schedule and dose management	Core	The system shall allow the patient to maintain medication schedules and shall create dose records for applicable active schedules and dates. Each generated occurrence shall start in PENDING state unless source-defined business logic	Valid schedules create expected occurrences; inactive schedules do not create future pending doses; repeated/concurrent generation does

			specifies otherwise.	not create duplicates.
FR-06	Dose outcome tracking	Core	The system shall permit the patient to record supported dose outcomes through validated, conditional state transitions. It shall retain saved dose events for history and analytics.	Invalid transitions are rejected or leave state unchanged; terminal/completed states are not overwritten by racing requests.
FR-07	Adherence history and analytics	Core	The system shall provide date-filtered dose history and summary analytics for a requested range. History shall support pagination with a default page size of 50 and a maximum of 100; analytics shall accept ranges up to 366 days.	Returned records match the authorized patient and date range; page limits and date-range limits are enforced; results use saved dose states.
FR-08	Care relationship invitation	Core	The patient shall be able to invite an existing account by exact email lookup and specify CAREGIVER or DOCTOR as relationship type. A new invitation shall create a PENDING relationship.	Valid invitation creates PENDING; unknown account or invalid relationship type is handled safely without exposing a searchable user directory.
FR-09	Accept or decline invitation	Core	The invited account shall be able to accept a PENDING relationship, changing it to ACTIVE, or decline it, changing it to REVOKED.	Only the invited user can act on their pending invitation; invalid state changes are rejected.
FR-10	Revoke consent	Core	The patient shall be able to revoke an ACTIVE relationship at any time. Revoked records shall remain for traceability; a later invitation shall create a new PENDING record rather than reactivate the old one.	After revocation, subsequent monitoring requests return no protected patient data; prior relationship record remains retained.
FR-11	Caregiver/doctor monitoring	Core	The system shall allow a caregiver or doctor to read only permitted monitoring data for a patient connected by an ACTIVE relationship of the matching role. Monitoring routes shall not permit changing patient dose status.	Correct active relationship permits read-only view; wrong role, missing, pending or revoked relationship is denied.
FR-12	Interaction check request	Supporting	The patient shall be able to request an interaction check by submitting medication identifiers. The backend shall verify that every identifier belongs to the authenticated patient before passing medication names to the configured provider. A mixed/foreign-ID request shall be rejected as a whole.	Owned identifiers reach provider adapter; any foreign identifier rejects the whole request; mock provider may return NO_DATA and must not imply safety.
FR-13	Notification centre	Supporting	The system shall list notifications belonging to the authenticated recipient and support pagination, unread-only filtering, unread count and read/unread updates.	Only recipient-owned notifications are returned or changed; another recipient's notification cannot be read or modified.
FR-14	Notification generation and deduplication	Supporting	The system shall create in-app notifications for supported events including missed-dose transitions, care-relationship changes and adherence attention. Duplicate notifications shall be prevented by recipient/type/event-key uniqueness. Care-team notifications shall be sent only when the matching relationship is ACTIVE at the relevant time.	Repeated processing does not create duplicate event notifications; inactive relationships do not receive care-team monitoring notifications.
FR-15	Health endpoint and consistent API errors	Supporting	The system shall expose a public process-health endpoint and return consistent success/error response envelopes. Error responses shall include a structured code and message and shall be paired with an appropriate HTTP status.	Health endpoint responds without authentication; API clients can distinguish success from failure using status and response body.

3.1 Dose Generation and Status Rules

- Generation shall run transactionally for a patient and calendar day, selecting active schedules valid for that day.
- Before inserting a dose occurrence, the system shall check for an existing occurrence. A database uniqueness constraint on (schedule, scheduled date, scheduled time) shall provide a second idempotency guard.
- Medication deactivation shall mark associated schedules inactive, mark pending doses CANCELLED, and preserve completed/historical dose records.
- CANCELLED doses shall be excluded from adherence analytics.
- Conditional state updates shall protect terminal states against concurrent overwrites.

The source documents do not provide a complete public state-transition table for every dose status. Therefore, this SRS requires validated and conditional transitions without inventing additional status names or transition permissions.

3.2 Adherence Calculation

For a requested date range, adherence is calculated from saved dose states:

Adherence (%) = 100 × (number of doses recorded as TAKEN) / (number of doses with a recorded outcome)

CANCELLED doses are excluded. The denominator includes only doses whose outcome has been recorded. The output is an application indicator, not a clinical measurement; it depends on complete and accurate user-entered events.

3.3 Adherence Attention Notification Rule

The project report specifies an application rule for month-to-date adherence attention: at least three recorded doses and a result below the configured 70% product threshold. This threshold is product logic only and is not a medically validated measure.

4. External Interface Requirements

4.1 User Interface

- The browser UI shall provide registration and login forms, patient medication/schedule forms, dose tracking views, date-filtered history and analytics.
- Patient views shall expose care-team invitation and consent-revocation actions.
- Caregiver and doctor dashboards shall visually distinguish their read-only monitoring role.
- The interaction checker shall clearly communicate that the submitted build uses a mock provider and that NO_DATA is not evidence of safety.
- Notifications shall support viewing, unread filtering/count and read/unread state changes.
- Validation and API errors shall be presented in a comprehensible manner without exposing sensitive internals.

4.2 Software Interfaces

Interface	Specification
Frontend ↔ Backend	REST API under /api/v1 using JSON request/response bodies.
Authentication	Bearer access token on protected requests; public exceptions include health, registration and login.
Database access	Spring Data JPA repositories using MySQL relational storage.
Schema management	Flyway migrations V1–V4 are authoritative; Hibernate validates the schema on startup.
Interaction provider	Provider abstraction; default implementation is a mock provider returning no clinical interaction data.
Monitoring	Spring Boot Actuator liveness/readiness endpoints are described in the project architecture.

4.3 Representative API Endpoints

Method	Endpoint	Purpose / access note
GET	/api/v1/health	Public process-health response.
POST	/api/v1/auth/register	Create a patient account.
POST	/api/v1/auth/login	Authenticate credentials and issue access token.
GET	/api/v1/users/me	Return current account profile; authenticated.
POST	/api/v1/interactions/check	Informational interaction request; authenticated and ownership-verified.
GET	/api/v1/notifications	List recipient-owned notifications; authenticated.
GET	/api/v1/relationships	List relevant care relationships; authenticated.

The supplied documents list representative endpoints rather than a complete endpoint catalogue. Additional medication, schedule, dose, analytics and relationship endpoints must follow the same versioning, validation and authorization conventions.

4.4 API Response and Error Format

Successful responses use a common envelope containing a success flag, an optional message and a data object. Error responses contain a failure flag and a structured error object with a code and message. INVALID_CREDENTIALS is one documented example. Clients shall use both HTTP status and response body rather than assuming that every response succeeded.

4.5 Communications

Deployed browser-to-API communication shall use HTTPS. API payloads shall use JSON. CORS origins and TLS/database connection settings shall be explicitly configured for the deployment environment.

5. Non-Functional Requirements

ID	Quality attribute	Verifiable requirement
NFR-01	Security	All protected endpoints shall require valid authentication except documented public endpoints.
NFR-02	Credential protection	Passwords shall be stored as BCrypt hashes; plaintext passwords and hashes shall not be included in profile or monitoring responses.
NFR-03	Authorization	Every protected data operation shall enforce server-side role and ownership rules. Client-supplied user IDs or role claims shall not override the authenticated principal/current database account.
NFR-04	Consent enforcement	Every care-team monitoring request shall re-check the current ACTIVE relationship and matching relationship type; revocation shall block subsequent requests.
NFR-05	Abuse resistance	Authentication and registration requests shall be rate-limited. The documented limiter is local to one backend instance; distributed deployment requires a shared gateway/store.
NFR-06	Data integrity	Foreign keys, unique constraints, transactions and conditional state updates shall protect relationships and dose/notification uniqueness.
NFR-07	History preservation	Medication/schedule deactivation shall preserve historical dose records and avoid rewriting completed dose outcomes.
NFR-08	Maintainability	Frontend, controller, service, repository, entity/DTO, security, configuration and exception concerns shall remain separated into clear modules.
NFR-09	Compatibility	The web application shall use standard REST/JSON and support modern browsers with JavaScript enabled.
NFR-10	Usability	Core tasks—registration, adding medication/schedule, recording a dose, viewing analytics and managing care consent—shall be reachable through clear navigation and date-oriented views.
NFR-11	Configurability	Database settings, JWT secret, CORS origins and timezone shall be configurable outside source code through environment/configuration settings.
NFR-12	Time correctness	Date calculations shall use an injected clock and configured IANA timezone, defaulting to Asia/Kolkata. Date-only values shall not be unnecessarily shifted through UTC.
NFR-13	Error handling	API failures shall use consistent structured error responses and suitable HTTP statuses; sensitive internal details shall not be exposed.
NFR-14	Observability	The backend shall expose documented liveness/readiness health endpoints.
NFR-15	Clinical honesty	The UI and API shall not represent application analytics as clinical measurements or a mock-provider NO_DATA response as proof of medication safety.

5.1 Performance and Capacity

The supplied documents do not state measured response-time, throughput, availability or concurrent-user targets. No numeric performance commitment is asserted in this SRS. Before production deployment, the project team should define targets and benchmark the system against a representative MySQL environment, particularly for concurrent dose generation and analytics.

5.2 Reliability and Recovery

The source documents describe persistence and integrity controls but do not establish a tested backup/recovery objective or production availability level. A production readiness plan should specify backup frequency, retention, restore testing, recovery-time objective and recovery-point objective before real patient data is accepted.

5.3 Privacy and Data Minimization

- Monitoring responses shall contain only data permitted by the active relationship and the monitoring purpose.
- Notifications shall be recipient-scoped.
- Passwords, password hashes and JWT secrets shall not be exposed in application responses.
- Production retention, access review, incident response and privacy impact assessment procedures remain operational requirements for future deployment.

6. Data Requirements and Integrity

6.1 Core Data Entities

Entity	Purpose	Principal relationships / constraints
User	Account identity, credentials hash and role.	One user owns many medications and notifications; stable identifier; normalized unique email.
Medication	Patient-owned medication information and active state.	Belongs to one owner; has one or more schedules; ownership checked on every relevant operation.
MedicationSchedule	Defines medication timing/frequency and active state.	Belongs to a medication; generates multiple dose records across dates.
DoseRecord	Scheduled occurrence, date, time and saved status.	Uniqueness on schedule + scheduled date + scheduled time; retained for history.
CareRelationship	Patient-to-care-team relationship, type and consent state.	State is PENDING, ACTIVE or REVOKED; role-matched ACTIVE relationship required for monitoring.
Notification	Application event addressed to one recipient, with read state.	Recipient-scoped access; uniqueness guard based on recipient/type/event key.

6.2 Data Integrity Rules

- Foreign keys shall prevent orphaned domain records.
- Email uniqueness shall be enforced after server-side normalization.
- Every medication/schedule/dose operation shall verify the authenticated owner's authority.
- Duplicate dose occurrences shall be prevented at both application and database levels.
- Duplicate notifications for the same recipient/type/event key shall be prevented.
- Flyway migration history shall define the authoritative schema version; Hibernate shall validate rather than silently mutate the schema.
- Revoked relationship records and historical dose records shall be retained for traceability/history.

6.3 Date and Time Rules

The backend shall use a configurable IANA timezone with Asia/Kolkata as the documented default. Calendar dates shall be stored and processed as date-only values where appropriate, avoiding unintended UTC conversion and off-by-one errors around midnight. Frontend and backend timezone configuration shall be consistent.

6.4 Retention and Deactivation

Deactivation is a soft state change. Related schedules become inactive, pending doses become CANCELLED, completed dose records remain unchanged, and historical records remain available for permitted history/analytics. CANCELLED events are excluded from the adherence calculation.

7. Security, Privacy and Consent Requirements

7.1 Authentication Rules

- Registration shall normalize and validate email and enforce uniqueness.
- Registration shall always assign PATIENT role; role values supplied by a public client shall be ignored.
- Login shall verify the password against the stored BCrypt hash and return a token only for valid credentials.
- Invalid login shall not disclose whether an email address exists.
- Each protected request shall validate token signature and expiry, reload the current account, and verify the token subject against the stable database identity.

7.2 Authorization Decision

Authorization shall combine three checks: (1) role suitability for the requested operation; (2) ownership of the requested resource; and (3) for care-team monitoring, an ACTIVE relationship of the matching CAREGIVER or DOCTOR type. Authorization shall be decided by the server using the current database account and relationship state, not by browser-supplied role or ownership claims.

7.3 Consent State Model

Current state	Event / actor	Next state	Required result
No relationship	Patient sends invitation	PENDING	Invitation recorded for an existing account.
PENDING	Invitee accepts	ACTIVE	Matching role may read permitted monitoring data.
PENDING	Invitee declines	REVOKED	No monitoring access granted.
ACTIVE	Patient revokes	REVOKED	Subsequent monitoring requests are denied.
REVOKED	Patient sends a new invitation	New PENDING record	Old record remains; it is not reactivated.

7.4 Client Token Storage and Risks

The project report documents sessionStorage for the access token. This limits persistence to the browser session but remains script-readable; a cross-site scripting flaw could expose it. The research paper identifies HttpOnly/SameSite cookies with CSRF protection or short-lived tokens with refresh rotation as possible hardening directions. These are future hardening options, not claims about the current build.

7.5 Security Limitations to Disclose

- The documented rate limiter is local to one backend instance.
- The research paper reports no audit logging or administrative tooling for trusted role provisioning.
- No formal security review or privacy impact assessment is reported.
- Integration tests use an in-memory H2 profile; MySQL-specific locking, constraint and date behaviour should be re-verified under concurrent load.

- Production use requires secure secret handling, TLS/database configuration, backups, access-review and incident-response procedures.

8. Use Cases and Main Workflows

UC-01: Register and sign in

Primary actor: Patient

Main flow: User opens the application, submits registration details, then logs in with valid credentials.

Postcondition: Account is created with PATIENT role; successful login returns access token and summary; invalid credentials return a generic error.

UC-02: Manage medication and schedule

Primary actor: Patient

Main flow: Patient adds a medication, configures a schedule and views generated dose occurrences.

Postcondition: Only the patient's own records are accessible; active schedule occurrences are generated without duplicates.

UC-03: Record dose outcome

Primary actor: Patient

Main flow: Patient opens a pending dose and records an available outcome.

Postcondition: Validated state transition is saved; historical state is retained; concurrent updates cannot silently overwrite protected terminal states.

UC-04: Review adherence

Primary actor: Patient

Main flow: Patient selects a date range and opens history/analytics.

Postcondition: Only own data is returned; range/pagination limits are enforced; cancelled doses are excluded from the calculation.

UC-05: Invite care-team member

Primary actor: Patient

Main flow: Patient enters an existing account's exact email and selects CAREGIVER or DOCTOR.

Postcondition: A PENDING relationship is created; no monitoring access exists until acceptance.

UC-06: Accept/decline invitation

Primary actor: Invitee

Main flow: Invitee reviews a pending invitation and accepts or declines.

Postcondition: Accept produces ACTIVE; decline produces REVOKED; only the invitee may act on the invitation.

UC-07: Monitor connected patient

Primary actor: Caregiver/Doctor

Main flow: Connected user opens dashboard for a patient.

Postcondition: Server validates token, current account, matching role and ACTIVE relationship before returning read-only permitted data.

UC-08: Revoke monitoring access

Primary actor: Patient

Main flow: Patient revokes an active relationship.

Postcondition: Relationship becomes REVOKED; subsequent monitoring requests are denied; historical relationship remains recorded.

UC-09: Check medication interactions

Primary actor: Patient

Main flow: Patient selects own medications and submits a check request.

Postcondition: All identifiers are ownership-verified; any foreign identifier rejects the entire request; mock NO_DATA is displayed with a clear disclaimer.

UC-10: Manage notifications

Primary actor: Any authenticated recipient

Main flow: Recipient lists notifications, filters unread items, checks unread count and marks an item read/unread.

Postcondition: Only recipient-owned notifications are returned or updated; duplicate event notifications are prevented.

9. Verification and Acceptance Test Plan

Verification shall combine unit tests, API/integration tests, authentication and authorization tests, Flyway migration tests, frontend API-client tests and browser-based acceptance tests. The following cases define expected outcomes; they are not a claim that a fresh test run has passed.

Test ID	Objective	Procedure / input	Expected result
TC-01	Register valid patient	Submit unique name/email/password.	One PATIENT account is created; response contains no password hash.
TC-02	Reject invalid authentication	Submit invalid credentials or access a protected route without a valid token.	Generic login error or protected-route denial; no sensitive information exposed.
TC-03	Reject cross-owner medication access	Patient A requests Patient B's medication by identifier.	Request is rejected and no foreign record is disclosed or changed.
TC-04	Reject unauthenticated access	Call a protected API without a valid token.	Request is denied.
TC-05	Enforce role and relationship	Caregiver/doctor requests a patient without matching ACTIVE relationship.	Monitoring data is not returned.
TC-06	Revoke consent immediately	Revoke ACTIVE relationship, then repeat monitoring request.	Next request is denied; revoked relationship remains stored.
TC-07	Check foreign medication IDs in interaction request	Submit a mixture of owned and foreign medication identifiers.	Whole request is rejected; provider is not given the foreign medication set.
TC-08	Handle interaction NO_DATA honestly	Use default mock provider.	UI/API indicates no data; does not label the combination safe.
TC-09	Scope notification update	Recipient A attempts to read/update Recipient B's notification.	Operation is rejected; Recipient B's record is unchanged.
TC-10	Prevent duplicate dose generation	Generate the same patient's scheduled doses for the same date repeatedly or concurrently.	At most one record exists for each schedule/date/time key.
TC-11	Preserve history on deactivation	Deactivate a medication with pending and completed doses.	Schedules become inactive, pending doses are CANCELLED, completed/history records remain.
TC-12	Verify analytics formula	Use a known set of taken, other recorded and	Percentage matches the documented formula; cancelled

		cancelled doses.	doses excluded.
TC-13	Validate pagination/range	Request page size above 100 or analytics range above 366 days.	Input is rejected or bounded according to documented API validation; no unbounded query is executed.
TC-14	Prevent self-assigned privileged role	Attempt public registration with role=DOCTOR or CAREGIVER.	Created public account remains PATIENT.
TC-15	Verify migration startup	Start application with schema at prior migration version.	Flyway applies ordered migrations and Hibernate validates resulting schema.
TC-16	Verify timezone boundary	Generate date-based doses around local midnight in configured timezone.	Dose dates follow configured calendar date without unintended UTC day shift.

9.1 Test Environment

- Backend tests: JUnit and Spring Boot Test; the documented test profile uses in-memory H2.
- Frontend: npm test script and API-client tests.
- Database-specific confirmation: execute migration, uniqueness, transaction and concurrency tests against MySQL 8 before production deployment.
- Manual acceptance: complete the end-to-end patient workflow and caregiver/doctor invitation, monitoring and revocation workflows in a browser.

9.2 Exit Criteria

- All Core requirements have corresponding passing tests in the target environment.
- All negative access-control tests demonstrate denial without data leakage.
- Repeated/concurrent dose generation does not create duplicate occurrences.
- Medication deactivation and relationship revocation preserve history while removing future/unauthorized access.
- Known limitations are visible in user-facing documentation and the interaction checker.
- Test results include environment, build/test commands, pass/fail counts and any unresolved defects; do not claim results without running tests.

9.3 Evaluation Evidence

The supplied research paper describes test categories and representative cases but explicitly does not report a fresh pass/fail count, code-coverage percentage, formal acceptance run, performance benchmark or user study. Such evidence must be collected separately if required by the evaluator.

10. Requirements Traceability Matrix

Requirement IDs	Subsystem / quality	Use case(s)	Verification cases
FR-01–FR-03	Authentication/profile	UC-01	TC-01, TC-02, TC-04, TC-14
FR-04	Medication ownership and deactivation	UC-02	TC-03, TC-11
FR-05–FR-06	Schedules and dose lifecycle	UC-02, UC-03	TC-10, TC-11, TC-16
FR-07	History and analytics	UC-04	TC-12, TC-13
FR-08–FR-10	Care relationship lifecycle	UC-05, UC-06, UC-08	TC-05, TC-06
FR-11	Read-only monitoring	UC-07	TC-05, TC-06
FR-12	Interaction check	UC-09	TC-07, TC-08
FR-13–FR-14	Notifications	UC-10	TC-09
FR-15	Health and error format	UC-01–UC-10	TC-02, TC-15
NFR-01–NFR-05	Security and consent	UC-01, UC-05–UC-09	TC-02–TC-09, TC-14
NFR-06–NFR-07	Integrity and retention	UC-02–UC-04	TC-10–TC-12, TC-15
NFR-08–NFR-15	Maintainability, usability, configuration, time and honesty	All relevant workflows	TC-08, TC-13, TC-15, TC-16

11. Assumptions, Dependencies and Constraints

11.1 Assumptions

- Users enter medication details and dose outcomes accurately and protect their credentials.
- Caregiver and doctor invitations target an existing registered account.
- The application is used for personal organization and academic demonstration rather than clinical decision-making.
- The backend and database are configured correctly before the application is used.
- Frontend and backend timezone settings are consistent.

11.2 Dependencies

- Java 17+, Spring Boot 3 dependencies and Maven wrapper for backend build/run.
- Node.js 18+, npm and the frontend package set for frontend build/run.
- MySQL 8 or a compatible configured database.
- Flyway migrations for schema initialization and evolution.
- A correctly configured JWT signing secret, CORS allowlist, database credentials and TLS settings for deployment.

11.3 Explicitly Unspecified Items

The supplied documents do not define exact response-time or throughput targets, a maximum supported user count, a complete dose-status transition matrix, a formal accessibility conformance level, a tested backup/recovery objective, or a full endpoint schema for every module. These items must be agreed and documented before treating them as acceptance commitments.

12. Known Limitations and Future Enhancements

Known limitation in the supplied project documents	Potential future work (not part of current guaranteed behaviour)
Mock interaction provider returns no clinical interaction data.	Integrate a validated/licensed interaction-data provider with documented data coverage and provenance.
No clinical validation; adherence values and 70% threshold are application logic.	Conduct ethically approved evaluation before making clinical or adherence-improvement claims.
No audit logging or administrative tooling for trusted role provisioning is reported.	Add audit trails and controlled administration for privileged role provisioning.
Rate limiting is local to one backend instance.	Use a shared gateway or distributed store for multi-instance deployments.
Token is stored in script-readable sessionStorage.	Evaluate HttpOnly/SameSite cookie protection with CSRF controls or short-lived token rotation.
H2 test behaviour may differ from MySQL under locking, constraints and dates.	Run database integration and concurrency tests against MySQL 8 and publish measured results.
No formal security review or privacy impact assessment is reported.	Complete a formal security review, privacy assessment, operational retention policy and incident-response plan.
Notifications are in-app; push/reminder channels are not described as implemented.	Add configurable reminder channels and explicit user preferences if required.
Accessibility and usability study are future work.	Perform accessibility review and representative-user usability testing.

Appendix A. Glossary

Term	Meaning in this SRS
ACTIVE relationship	Accepted care relationship that permits role-matched read-only monitoring.
PENDING relationship	Invitation awaiting the invitee's decision; grants no monitoring access.

REVOKED relationship	Declined or revoked relationship; grants no monitoring access and remains in history.
Idempotency	Repeated execution of the same dose-generation request does not create duplicate records.
Read-only monitoring	Viewing permitted data without permission to change the patient's dose status or medication records.
Soft deactivation	Disabling future use while preserving relevant historical records.
Acceptance test	A test with a defined input/action and observable expected result.

Appendix B. Source Documents and Specification Notes

This SRS was consolidated from the two user-provided PDFs: (1) SmartMed_IEEE_Research_Paper.pdf and (2) NEW SMARTMED project report.pdf. The research paper provides architectural/security rationale, verification strategy, known threats to validity and future work. The project report provides the academic project context, module descriptions, user workflows, hardware/software requirements and representative API routes.

Specification discipline: this document distinguishes current documented behaviour from future recommendations. It does not state that the application has been clinically validated, production-deployed, performance-benchmarked or freshly tested during preparation of this SRS.

End of Document.